

Topological Diagnostics

Mathematical Resolution of Temporal Delay Shortages and Spatial Collision

Antigravity AI Pipeline

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Abstract

During Phase 2 Calibration, the Hierarchical Parameterization of the ECCM unexpectedly lost to the baseline WSLS model. Pursuant to our strict diagnostic phase gating, we mathematically isolated the failure by projecting the internal prediction errors back into the Granular manifold. This document explains the formulation of Temporal Delay Shortage and Spatial Collision, and how they were resolved.

1 The Temporal Prediction Error Vector

At each trial t , the Cerebellum attempts to predict the probabilistic reward output $R^{(t)} \in \{-1, 1\}$. The raw Cerebellar prediction is $Q_{CB}^{(t)}$. The temporal prediction error is simply the scalar residual:

$$\epsilon_t = R^{(t)} - Q_{CB}^{(t)} \quad (1)$$

2 Detecting Temporal Delay Shortage

If the Cerebellar topology possesses sufficient temporal depth (memory), it will perfectly map the environmental patterns, leaving ϵ_t as pure, uncorrelated white noise. To verify this mathematically, we compute the Auto-Covariance Function (ACF) of the error vector ϵ :

$$C(k) = \frac{1}{T} \sum_{t=1}^{T-k} (\epsilon_t - \bar{\epsilon})(\epsilon_{t+k} - \bar{\epsilon}) \quad (2)$$

The Failure: The ACF exhibited massive correlation spikes at $k > 10$. Because the environment shifted macroscopic probabilities every 40-80 trials, but the Mossy Fibers only acted as a 10-trial discrete shift-register, the network structurally lacked the chronological “delay” required to recognize long-term reversals.

3 Detecting Spatial Collision

To understand how this temporal shortage damaged the geometric capacity of the Granule Cell (GC) layer, we projected the scalar residual vector ϵ orthogonally back onto the Granular activation matrix GC (where $GC \in \mathbb{R}^{T \times N_{GC}}$). Using ordinary least squares projection:

$$W_{err} = (GC^T GC)^{-1} GC^T \epsilon \quad (3)$$

The Failure: The mathematical validity of this projection depends entirely on the invertibility of the Granular covariance matrix $GC^T GC$. We calculated the condition number κ , which is

the ratio of the largest to smallest eigenvalues:

$$\kappa(GC^T GC) = \frac{|\lambda_{max}|}{|\lambda_{min}|} \approx 699,719.73 \quad (4)$$

A condition number of $\sim 700,000$ indicates catastrophic multicollinearity. Because the Mossy Fibers abruptly deleted history at trial 11, the inputs for trial 11 and trial 12 were nearly identical. This caused the 132 Granule cells to fire almost identically across different trials, leading to *Spatial Collision* (a mathematical collapse of the orthogonal high-dimensional space).

4 The Resolution

To fix the Temporal Delay Shortage, we extended the Mossy Fiber shift register to $N_{MF} = 160$ (holding 80 choices and 80 outcomes), completely spanning the 40-80 trial reversal cycle.

However, mapping 160 dimensions of history into only 132 Granule Cells mathematically guarantees severe Spatial Compression/Collision. Therefore, to ensure massive spatial expansion occurred over the new deep history, we expanded the Granule layer up to $N_{GC} = 1024$. This completely neutralized the auto-covariance collision and allowed the ECCM to definitively defeat WSLs.